THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Course: Parallel and Distributed Computing

High-Performance Grid-Based Environmental Spread Modeling Using Multithreaded CPU Parallelism

Project Synopsis

Parallel and Distributed Computing is essential for accelerating large-scale computational simulations. Many natural phenomena, such as heat diffusion and forest fire propagation, can be modeled using grid-based numerical techniques where each cell updates its state based on neighboring values over discrete time steps.

This project presents the design and implementation of a parallel simulation framework using **C++ and OpenMP** to model spatial diffusion and combustion dynamics on a two-dimensional grid. The environment is discretized into cells representing temperature values or fire states (empty, vegetation, burning). At each iteration, cell states are updated according to predefined mathematical or probabilistic rules.

Since cell updates within a time step are independent, OpenMP is used to parallelize computations across multiple CPU cores. The system compares sequential and parallel implementations to evaluate execution time, speedup, and scalability under varying grid sizes and thread counts.

The objective of this project is to demonstrate performance improvement through shared-memory parallel programming and to analyze scalability behavior in grid-based simulations.